

Predicting Childhood Stunting Risk from Field-Recordable Survey Variables Using Decision Tree Learning

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Abstract—Early identification of children at risk of stunting is critical for timely intervention. This study presents an interpretable decision tree that classifies stunting risk into low, moderate and high levels using nine field-recordable variables from the 2023 Indonesian Health Survey (SKI) microdata, covering 304,193 children aged 0 to 59 months. Risk labels follow the height-for-age Z-score (HAZ) of the WHO Child Growth Standards. On a stratified hold-out set of 60,839 children the model attains 0.7557 accuracy, macro F1 0.7373 and weighted F1 0.7468, with the highest per-class F1 for the low-risk class (0.8195) and the lowest for the moderate-risk class (0.6656), which combines high precision (0.8528) with low recall (0.5457). As a binary screen, in which the at-risk band aggregates the moderate and high classes, it reaches 73.97% sensitivity, 92.59% specificity and 92.76% positive predictive value. Importance is dominated by measured height (0.5243) and child age (0.4580), the very quantities from which HAZ is computed, whereas the six maternal, environmental and socioeconomic predictors together contribute below 0.0002. The model is therefore characterised not as an aetiological predictor of stunting but as a field-deployable approximation of the WHO height-for-age decision boundary, executable at the Posyandu without LMS lookup tables. Ensemble learners reach higher accuracy on the same protocol, 0.9461 for gradient boosting and 0.9413 for Random Forest, at the cost of hand-executability, while a model trained without anthropometric inputs raises macro F1 only from 0.2031 to 0.3565, which bounds the standalone contribution of the non-anthropometric factors. The rules are integrated with actionable recommendations in the GENY-StuntCare application.

Keywords—decision tree, stunting screening, height-for-age Z-score, interpretable machine learning, SKI 2023 microdata, screening sensitivity

I. INTRODUCTION

Stunting remains a major public health issue in child development [1]. The World Health Organization defines it as impaired growth from poor nutrition, repeated infection and inadequate stimulation in early life, and classifies a child as stunted when the height-for-age Z-score falls below minus two standard deviations of the WHO Child Growth Standards [2]. Consequences extend beyond physical growth to cognition, schooling, productivity and chronic disease risk [3].

Although efforts to eradicate child malnutrition have been made worldwide [3], prevalence in many developing countries, including Indonesia, remains high. SKI 2023 reported national stunting prevalence near 21.5% among children under five, above the national target of 14% [4]. The persistence of stunting reflects determinants spanning maternal health, socioeconomic conditions, sanitation, nutritional intake and access to care [5].

Machine learning has been applied to this problem because it exposes patterns that classical statistics do not readily reveal [6], using Random Forest [7], Support Vector Machine [8] and gradient boosting or stacking ensembles [9]. The decision tree is attractive here because it models the decision process as a hierarchy of single-attribute tests [10], so practitioners can inspect which variables drive a risk assignment [5], and because it is computationally inexpensive. Few studies, however, examine how one transparent tree behaves when anthropometric, maternal, environmental and socioeconomic fields available at the Posyandu are combined in a single pipeline.

This study builds such a system on SKI 2023 microdata. The contribution is threefold. A fully specified pipeline over 304,193 children is reported with complete disclosure of merge coverage, missing-data rates and deduplication. The relative contribution of anthropometric and non-anthropometric predictors is quantified rather than asserted. The model is evaluated as a screening instrument through sensitivity, specificity and predictive values, and benchmarked against ensemble alternatives so that the interpretability and accuracy trade-off is explicit.

II. RELATED WORK

A. Machine Learning for Stunting in Indonesia

Benchmarking studies on national-style data report similar accuracy ceilings across algorithms, suggesting the signal is bounded by available variables rather than model sophistication [11]. A meta-analysis of machine learning on demographic and health survey data places pooled accuracy for child-malnutrition models near 68.9% [12]. Others improve performance with stacking and resampling for class imbalance [9], while a systematic review notes that this literature reuses a narrow variable set and rarely emphasizes

interpretability [6]. A parallel line pairs fuzzy inference with expert-system reasoning [13], and metaheuristic tuning of fuzzy systems is well established [14]. The present study is complementary: rather than maximizing accuracy with an opaque model, it asks how far one transparent tree can go while remaining explainable.

B. Interpretability and Decision-Oriented Evaluation

The value of a screening model depends on how it is used. The predictive strength of an individual flag weakens where effective prevention already exists, because intervention removes the outcomes the model anticipates [15], [16]. Classical epidemiology further holds that factors distinguishing high-risk individuals differ from those that shift population incidence, so a model that appears modest individually can still guide population action [17]. For deployment, methods weighing false-positive cost against benefit, such as decision curve analysis, are more informative than accuracy alone [18], [19].

C. Height-for-Age Z-Score (HAZ) Standard

HAZ expresses a child height relative to the median of a healthy reference population of the same age and sex, scaled by the reference standard deviation [20]. A child is stunted below minus two standard deviations and severely stunted below minus three [21]. Table I gives the three risk bands used throughout.

HAZ serves as the ground truth for the label, not as a model input. Because HAZ is by construction a deterministic function of height, age and sex, and all three are present in the predictor set, the label and part of the feature space are analytically linked. Height and age are nevertheless retained because they are the fields Posyandu cadres already record, so the model learns the WHO classification pattern from data genuinely available at screening time; the consequence is developed in Section IV.D. Concurrent body weight is excluded to avoid a second co-measured anthropometric channel.

TABLE I. HAZ-BASED DEFINITION OF THE RISK CLASSES

Risk level	HAZ range	Interpretation
Low	$\text{HAZ} \geq -1$	Normal height-for-age
Moderate	$-2 \leq \text{HAZ} < -1$	At risk of stunting
High	$\text{HAZ} < -2$	Stunted or severely stunted

III. METHODS

A. Medical numerical data preprocessing

Medical data are difficult to analyse: interpretation differs between surveys, missing values are frequent, and out-of-range codes are common, because each condition has its own instrument, target population and symptom set [22]. Raw records also carry no analytic label, so labelling precedes classification. A further constraint is that imputation of numerical medical data cannot be performed carelessly, because every record corresponds to an actual patient condition; a value plausible in aggregate may be clinically implausible for an individual child, so substitution rules must be applied per variable rather than uniformly across all numeric fields [23].

The SKI 2023 files separate child, individual and household records and are merged on the household key. Numeric fields are then parsed from comma-decimal notation, categorical fields are decoded into discrete codes, and survey non-response codes are replaced with missing per variable

following the codebook: 88 for gestational age and 888 or 8888 for birth weight, while measured height and child age carry no non-response code. Empty numeric entries are imputed with the column median and empty categorical entries mapped to an explicit missing category. Records whose nine-dimensional predictor vector duplicates an earlier record are removed, and the three-class label is finally derived from HAZ.

The 2023 SKI data, split into several Socioeconomic and Nutritional feature files, will be merged by household ID. Next, we normalize the data and decode the categorical value as a code number representing each category. Next, we handle missing values. After all the data is complete, we select the important values that affect the prediction before performing advanced labeling.

B. Decision Tree

A decision tree recursively partitions the data into increasingly homogeneous subsets, selecting at each node the attribute that best separates the classes; in the ID3 and C4.5 family this is guided by entropy and information gain [24]. For a set S with c classes and class proportions p_i , entropy is given in (1) and the information gain of attribute A in (2), where S_v is the subset of S for which A takes value v .

$$H(S) = - \sum_{i=1}^c p_i \log_2 p_i \quad (1)$$

$$IG(S, A) = H(S) - \sum_v (|S_v|/|S|) H(S_v) \quad (2)$$

The attribute with the largest gain is selected and the procedure repeats until a stopping criterion is met, yielding attribute tests at interior nodes and class decisions at leaves. Each prediction is therefore a short chain of conditions, which is the property that makes the model interpretable for health workers.

C. Dataset

The data are SKI 2023 microdata released by the Ministry of Health [22]. Table II reports the construction of the analytic dataset. The child file contributes 306,281 records with valid HAZ, mean -1.131 and standard deviation 1.325 . The household file merges completely, whereas the individual file merges for 75.2% of children because the linkage selects the spouse of the household head, absent in about one quarter of households. After removing 2,088 duplicate predictor vectors, 304,193 records remain.

TABLE II. DATASET CONSTRUCTION AND CLASS DISTRIBUTION

Stage	Value
Child file, 0–59 months	$306,281 \times 33$
Individual file	$877,531 \times 413$
Household file	$315,646 \times 16$
Records with valid HAZ	$306,281$ (100.0%)
HAZ, mean $\pm$ SD	-1.131 ± 1.325
Merge coverage, individual file	$230,405$ (75.2%)
Duplicate vectors removed	$2,088$ (0.68%)
Final analytic dataset	$304,193$
Low / Moderate / High	$133,269 / 102,743 / 68,181$
Class share	$43.8\% / 33.8\% / 22.4\%$
Train / test, stratified 80/20	$243,354 / 60,839$

Class proportions describe the unweighted analytic sample; survey weights were not applied.

Nine predictors were retained. Table III gives their source, type and missing-data rate.

TABLE III. PREDICTORS, SOURCES AND MISSING-DATA RATES

Predictor	Source	Type	Missing
jenis_kelamin	B4K4 · balita	Cat, 2	0.0%
usia_bulan	B4K7BLN · balita	Num	0.0%
berat_badan_lahir	I05A · balita	Num	5.3%
tinggi_badan	J02B · balita	Num	0.5%
usia_kehamilan	I04 · balita	Num	3.8%
klasifikasi_wilayah	B1R5 · balita	Cat, 2	0.0%
maternal_pulmonary_TB	A12 · individu	Cat, 3	24.8%
sumber_air	B6R1 · rumahtangga	Cat, 13	0.0%
sanitasi	B6R11 · rumahtangga	Cat, 5	0.0%

Numeric gaps are median-imputed (3,100 g; 88.10 cm; 39.0 weeks); categorical gaps take an explicit missing level. A12 records pulmonary tuberculosis diagnosis in the past year (1 yes within six months, 2 yes earlier, 3 no); it is a maternal health indicator, not a measure of maternal nutritional status.

D. Proposed Method

This study proposes a data processing flow using several preprocessing methods while simultaneously using a decision tree as a classification method.

The pipeline of Section III.A feeds a single decision tree, fitted on a stratified 80/20 split with random state 42. The classifier uses the entropy criterion, maximum depth eight, minimum 50 samples to split, minimum 20 samples per leaf and balanced class weights. The fitted tree realises depth eight with 179 leaves, comprising 65 low, 70 moderate and 44 high. The model and its feature metadata are serialised so that the inference program consumes exactly the training schema.

TABLE IV. DATA PREPROCESSING PIPELINES

Step	Operation
Data merger	Join balita, individu, and rumahtangga on the mother/household key (IDRT)
Key merger	Cast IDRT to a common format so records align across files.
Decimal parsing	Parse Indonesian comma-decimal numeric fields correctly to avoid silent corruption.
Data type handling	Keep numeric features numeric; encode categorical features as discrete codes.
Missing-code substitution	Replace survey non-response codes with missing, applied per variable following the SKI codebook: 88 for gestational age, 888 or 8888 for birth weight. Measured height and child age carry no non-response code.
Missing value handling	Map empty entries to an explicit missing category via a safe ordinal encoder.
Deduplication	Remove records whose nine-dimensional predictor vector duplicates an earlier record.
Feature selection	Use nine predictors derived from the merged child, maternal and household records.
Labeling	Derive the three-class risk label from the HAZ value (WHO standards).

Next, the data will be divided into 80% training and 20% testing data for classification using a decision tree. The decision tree settings can be seen in Table V.

TABLE V. DECISION TREE PARAMETERS

Parameter	Value
Split criterion	entropy (information gain)
Maximum depth	8 (actual depth 8, 179 leaves)
Minimum samples to split	50
Minimum samples per leaf	20
Class Weight	balanced
Train / test split	stratified (80/20)
Random state	42

For reproducibility, the trained model and its feature metadata (feature_columns.json and feature_types.json) are serialized with joblib so that the inference program consumes exactly the schema produced during training.

IV. RESULT

A. Preprocessing result

The 2023 SKI dataset consists of three main files: "balita" (representing features associated with toddlers), "individu" (representing each woman or mother), and "rumahtangga" (household) (representing the social status of households). The data was merged into one large dataset based on household IDs.

Two properties of the merge deserve comment. The household file aligns completely with the child file whereas the individual file aligns for 75.2% of children, so the single maternal predictor rests on a partial linkage. Deduplication removes 2,088 records distributed as 841 low, 927 moderate and 320 high, so the moderate class loses the largest proportional share at 0.89%.

B. Decision tree result

The classifier was trained on 243,354 records and evaluated on the held-out 60,839. Table VI reports per-class results and Table VII the confusion matrix as counts, so that every quantity in Table VI can be recomputed independently.

TABLE VI. TESTING MODEL ACCURACY

Class	Precision	Recall	F1	Support
Low	0.7350	0.9259	0.8195	26,654
Moderate	0.8528	0.5457	0.6656	20,549
High	0.7145	0.7394	0.7268	13,636
Accuracy	—	—	0.7557	60,839
Macro avg	0.7674	0.7370	0.7373	60,839
Weighted avg	0.7702	0.7557	0.7468	60,839

TABLE VII. CONFUSION MATRIX, ROWS ACTUAL, COLUMNS PREDICTED

Actual \ Pred.	Low	Moderate	High	Total
Low	24,680	1,354	620	26,654
Moderate	5,927	11,214	3,408	20,549
High	2,972	581	10,083	13,636
Total	33,579	13,149	14,111	60,839

Accuracy of 0.7557 must be read with the class distribution: the low class holds 43.8% of the sample, so a majority-class classifier would reach 0.4381 and the model discriminates well above the base rate. The narrow gap between macro F1 (0.7373) and weighted F1 (0.7468) indicates the model does not collapse onto the majority class.

C. Error Analysis and Screening Performance

A moderate-class recall of 0.5457 has direct consequences for intervention allocation. Of 20,549 true moderate cases, 9,335 are misclassified, 5,927 into the low band and 3,408 into the high band, so the moderate band leaks in both directions, consistent with its position between the -2 and -1 thresholds. The low class is most stable, with 620 cases escalated to high and 1,354 to moderate. For the high class the dominant error is a two-level displacement: 2,972 high-risk children fall into the low band against 581 into the moderate band.

This asymmetry is structural. The moderate band occupies the narrowest HAZ interval, and a depth-limited tree approximates the height-for-age surface with piecewise-constant regions, so the narrowest band absorbs the largest discretisation error; Section IV.D shows 83.8% of leaves lie at the depth limit. In deployment the model acts as a binary triage instrument in which the at-risk band aggregates moderate and high and drives referral, so Table VIII reports the quantities that govern that decision.

TABLE VIII. PERFORMANCE AS A BINARY SCREEN (AT-RISK = MODERATE OR HIGH)

Quantity	Value	Ratio
Sensitivity	73.97%	25,286 / 34,185
Specificity	92.59%	24,680 / 26,654
Positive predictive value	92.76%	25,286 / 27,260
Negative predictive value	73.50%	24,680 / 33,579
High-risk children cleared	21.80%	2,972 / 13,636
Low-risk children flagged	7.41%	1,974 / 26,654

A positive predictive value of 92.76% indicates that cadre effort spent on flagged children is rarely misdirected, whereas a sensitivity of 73.97% is the principal operational weakness. Threshold tuning and cost-sensitive learning should therefore target sensitivity rather than overall accuracy.

D. Decision Rules and Feature Importance

The principal virtue of the model is that its reasoning can be inspected. The initial split is on measured height, and the tree thereafter alternates between measured height and child age, with sex and the remaining predictors appearing only marginally. Because height and age are exactly the quantities from which HAZ is computed, the tree effectively re-derives the WHO height-for-age boundary from field measurements. Table IX quantifies this two ways, by impurity-based importance and by the number of split nodes in which each predictor is used.

TABLE IX. FEATURE IMPORTANCE AND STRUCTURAL USE IN THE TREE

Predictor	Importance	Splits	Share
tinggi_badan	0.5243	73	41.01%
usia_bulan	0.4580	63	35.39%
jenis_kelamin	0.0175	26	14.61%
berat_badan_lahir	< 0.00005	5	2.81%
usia_kehamilan	0.0001	3	1.69%
klasifikasi_wilayah	< 0.00005	3	1.69%
maternal_pulmonary_TB	< 0.00005	2	1.12%
sumber_air	< 0.00005	2	1.12%
sanitasi	< 0.00005	1	0.56%
Total	1.0000	178	100.00%

All nine predictors have strictly positive importance and appear as split nodes; six contribute below 0.00005 each and are reported as such rather than rounded to zero. The tree holds 178 internal nodes against 179 leaves, as a binary tree requires.

Three structural observations follow. First, 150 of 179 leaves (83.8%) lie at the maximum depth of eight, so the partition was halted rather than completed, as expected when a piecewise-constant model approximates a smooth surface. Second, of 83 sibling leaf pairs, 64 (77.1%) assign the same class to both children, so most terminal splits reduce entropy without changing any prediction and could be merged by post-pruning without loss. Third, the six non-anthropometric predictors are used only in deep, low-support regions.

Fig. 1 presents the tree to depth three. The first four levels alternate exclusively between measured height and child age, which is precisely the two-dimensional height-by-age lattice on which the WHO reference tables are defined.

```

tinggi_badan <= 99.95
|- usia_bulan <= 47.50
| |- tinggi_badan <= 93.95 -> usia (36.50)
| +- tinggi_badan > 93.95 -> usia (40.50)
  +- usia_bulan > 47.50
    |- tinggi_badan <= 95.65 -> TB (94.35)
    +- tinggi_badan > 95.65 -> usia (52.50)
      tinggi_badan > 99.95
      |- tinggi_badan <= 104.45 ---> usia (52.50)
      +- tinggi_badan > 104.45 ---> TB (105.25)

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Fig. 1. Trained decision tree to depth three; the value in parentheses is the next split threshold.

The concentration of importance on growth measurements indicates that the current predictor set is anchored on

anthropometry [1], [25]. The interpretation is therefore stated explicitly: within this feature set the model does not identify the causes of stunting, it reconstructs the WHO height-for-age decision boundary from inputs a cadre can record. That is operationally useful, since it removes the need for LMS lookup tables in the field, but it is not evidence of aetiological prediction. Complementary-feeding timing likewise shows only weak association with stunting in the WHO evidence review, so feeding signals are best read as markers of overall feeding adequacy [26].

E. Benchmark and Model Without Anthropometric Inputs

To make the interpretability and accuracy trade-off explicit, the tree is benchmarked against three alternative learners under an identical protocol: the same dataset, the same stratified 80/20 partition and a fixed random state of 42, with a majority-class predictor as lower bound. Because every learner on this feature set exploits the same anthropometric channel, a benchmark alone cannot separate anthropometric from non-anthropometric signal. A further model is therefore trained on the same records and labels with measured height and child age withheld, so the contribution of the remaining factors is measured directly rather than inferred from an importance ranking.

Three changes apply to that model. Mid-upper arm circumference from the individual file serves as the maternal nutrition indicator, dichotomised at the national chronic energy deficiency threshold of 23.5 cm [4]; it is available for 186,921 linked mothers, of whom 6.7% fall below the threshold. Drinking-water source is one-hot encoded, because ordinal coding of its thirteen levels orders them lexicographically and does not preserve the gradient of water-source adequacy. Birth weight and gestational age are additionally represented as low-birth-weight and preterm indicators. Maternal records are linked through the explicit mother identifier rather than the household-head spouse relation.

TABLE X. BENCHMARK UNDER AN IDENTICAL PROTOCOL

Model	Acc.	Macro F1	Recall high	By hand
Majority baseline	0.4381	0.2031	0.0000	Yes
Decision tree (proposed)	0.7557	0.7373	0.7394	Yes
Random Forest	0.9413	0.9394	0.9306	No
Gradient boosting	0.9461	0.9444	0.9592	No
Logistic regression	0.6859	0.6742	0.7056	Partly
Without height and age	0.3938	0.3565	0.5133	Yes

The two ensemble learners outperform the proposed tree by a wide margin, with gradient boosting reaching 0.9461 accuracy and Random Forest 0.9413 against 0.7557. The gap is consistent with the interpretation advanced in Section IV.D: because the label is a deterministic function of height, age and sex, greater model capacity yields a finer approximation of the underlying height-for-age surface rather than new epidemiological insight. Logistic regression, restricted to a linear decision boundary, reaches only 0.6859, which confirms that the boundary is non-linear in height and age. The trade-off is therefore explicit. A single depth-eight tree with 179 leaves can be printed on one page and applied by hand at the Posyandu, whereas a 200-tree forest cannot be inspected or executed without a computing device. The accuracy the tree forgoes buys auditability, and in a setting where cadres must justify a referral to a caregiver that property is not incidental.

The model trained without measured height and child age reaches 0.3938 accuracy, below the 0.4381 majority baseline, but this comparison is not like for like: the model applies

balanced class weights whereas the baseline does not, so it trades majority-class accuracy for minority recall. On the balanced criterion the contribution is positive and modest, with macro F1 rising from 0.2031 to 0.3565 and recall on the high-risk class from 0.0000 to 0.5133. Non-anthropometric factors therefore carry real but weak signal, and the anthropometric channel accounts for a further 0.3808 of macro F1.

F. Implications for Low-Resource Deployment

A tree of modest depth runs on inexpensive hardware and, once pruned, can be applied by hand, reducing each prediction to a short chain of questions suited to low-effort screening. The system is deployed as GENY-StuntCare and piloted as a mobile minimum viable product with a community partner in Sidoarjo, East Java, where cadres enter the nine fields through a guided form and the returned band drives feeding modification or referral. Transparency also supports trust and allows supervisors to audit the logic. Two constraints govern the correspondence between form and model: the form must expose all thirteen drinking-water categories, and the maternal field must expose the three response levels of the tuberculosis item on which the model was trained. Records collected through the application are retained for prospective validation and form no part of the training set reported here.

G. Limitations

Construct validity is the principal limitation. HAZ is a deterministic function of height, age and sex, all present in the predictor set, so much of the discriminative power is analytic rather than epidemiological. An accuracy of 0.7557 should be read as the fidelity with which a depth-eight partition approximates the WHO height-for-age surface, not as prediction of stunting from independent risk factors.

Four preprocessing decisions affect the reported figures. Non-response codes were substituted uniformly across numeric predictors, whereas the codebook defines 88 as a non-response code only for gestational age; measured height carries no such code, so genuine heights of 88.0 cm became missing and were imputed to the median of 88.10 cm, affecting the densest region of the dominant predictor, and the 0.5% missing rate for height is an upper bound. Records missing height were imputed rather than dropped, yielding feature and label pairs inconsistent by construction. Median values were computed before partitioning, so a small amount of test-partition information enters the training features. Deduplication ignored the label, collapsing records with identical predictors but differing labels, which removes irreducible Bayes error and biases metrics optimistically; the affected volume is 2,088 records, or 0.68%. All four are addressed in the model of Section IV.E.

Representation is also imperfect. Drinking-water source has thirteen levels encoded ordinally after lexicographic sorting, placing surface water adjacent to bottled water, so its near-zero importance reflects encoding as much as absence of signal. The maternal channel is weak on three counts: the variable records tuberculosis rather than nutrition, it is missing for 24.8% of the sample, and the linkage selects the spouse of the household head rather than the mother identified by the explicit mother key, excluding mothers who are themselves household heads. The sample is also unweighted, so class proportions describe the analytic sample rather than the weighted national population, which is why the high-risk

share of 22.4% differs from the weighted national prevalence of 21.5% [4].

Sensitivity remains the operational weakness: in screening the minority high-risk class matters most, and precision degrades further as the base rate falls [27]. Height-for-age labels should also be read with care, since apparent change can arise from measurement and metric effects rather than real recovery [28].

V. CONCLUSION

This study presents an interpretable decision tree classifying stunting risk into low, moderate and high levels from nine field-recordable predictors in SKI 2023 microdata. On 60,839 held-out children it reaches 0.7557 accuracy, macro F1 0.7373 and weighted F1 0.7468, with per-class F1 highest for the low class (0.8195) and lowest for the moderate class (0.6656), the latter reflecting the narrowness of the moderate HAZ band. As a binary screen it achieves 92.76% positive predictive value against 73.97% sensitivity, identifying sensitivity as the target for threshold tuning.

The principal analytic finding concerns the composition of the predictive signal: measured height and child age account for 0.9823 of total importance, and these are precisely the quantities from which HAZ is computed. Within this predictor set the model is therefore best understood as a field-deployable approximation of the WHO height-for-age decision boundary rather than an aetiological model, and the model of Section IV.E quantifies how much risk signal the non-anthropometric factors carry alone. The tree is small enough for resource-constrained deployment and its branches translate directly into the recommendations delivered by GENY-StuntCare. Future work will apply survey weights, correct per-variable non-response handling, link mothers through the explicit key, replace ordinal coding, calibrate thresholds against intervention costs, and validate prospectively in the field.

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